THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–12) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A person eats only steamed vegetables and fruits and nothing else. The likely lack is: 1

(a) water **(b)** fibre **(c)** vitamins **(d)** protein and fat

2. A mixture contains aluminium, iron and wooden pieces. A magnet separates: 1

(a) iron and aluminium **(b)** iron only
(c) wood **(d)** all metals

3. Which of the following is true about a freely falling stone? 1

(a) it moves in a circle
(b) it is at rest
(c) it is in periodic motion
(d) it is in linear motion and its speed increases

4. If material C is an opaque metal and D is opaque wood, how would you distinguish them? 1

(a) C conducts heat and D does not **(b)** both float
(c) both transparent **(d)** both dissolve

5. If a 20 °C object touches a 60 °C object, which gets warmer? 1

(a) the 60 °C object **(b)** the 20 °C object **(c)** both cool **(d)** neither

6. If 50 mL of water evaporates from a vessel in 5 days, the average loss per day is: 1

(a) 10 mL **(b)** 50 mL **(c)** 5 mL **(d)** 250 mL

7. A saturated solution of salt at 30 °C contains 36 g salt in 100 g water. In 50 g water, the maximum salt that dissolves is: 1

(a) 36 g **(b)** 9 g **(c)** 72 g **(d)** 18 g

8. Two groups of seeds are kept: Group A in water, group B dry. Only A sprouts. What is the conclusion? 1

(a) dry seeds are dead **(b)** seeds do not need water
(c) water is needed for germination **(d)** sprouting is random

9. Earthworms help soil by: 1

(a) loosening and mixing it **(b)** removing humus
(c) polluting it **(d)** making it hard

10. The weight of the same object on the Moon compared with Earth is: 1

(a) one sixth **(b)** six times **(c)** equal **(d)** zero

11. Which of the following is not good practice while recording data? 1

(a) giving a title
(b) rounding off numbers to look neat without noting it
(c) writing units
(d) noting the date

12. A jungle has more kinds of organisms than a desert because: 1

(a) it is colder **(b)** it has no plants
(c) it has no animals **(d)** it provides more food, water and shelter

13. An experiment gave these results: Food X – blue-black with iodine; Food Y – oily patch; Food Z – no change in either test. Which is correct? 1

(a) X has starch, Y has fat, Z has neither starch nor fat
(b) X has fat, Y has starch, Z has both
(c) X has no starch, Y has starch, Z has fat
(d) all three contain starch

14. A student suspends magnet A and brings magnet B's north pole near A's north-seeking end. A turns away. This shows: 1

(a) B's north attracts A's north
(b) B has no poles
(c) B's north repels A's north
(d) A is iron

15. A student measures a pencil three times and gets 8.3 cm, 8.2 cm and 8.4 cm. The best value to report is: 1

(a) 8.4 cm **(b)** 8.3 cm **(c)** 8.2 cm **(d)** 24.9 cm

16. Which of the following is a reason to use thermocol for packing glass items? 1

(a) it is light and cushions shocks
(b) it is a metal
(c) it is hard
(d) it is transparent

17. A student heats a metal rod and a glass rod at one end for 1 minute. She touches the other ends. She will find: 1

(a) metal rod hotter **(b)** no change **(c)** both same **(d)** glass rod hotter

18. Why does a puddle disappear faster than a lake in the same sun? 1

(a) lake is salty
(b) puddle is hotter
(c) smaller volume and large surface to depth ratio
(d) lake has no water

19. Why does purifying water require both filtration and chlorination or boiling? 1

(a) filtration removes salt
(b) boiling removes sand
(c) filtration cannot remove all germs
(d) sedimentation removes germs

20. A tadpole's tail disappears as it grows. This is an example of: 1

(a) response to light
(b) death
(c) change in body during growth
(d) reproduction

21. Which of the following resource-use decisions is sustainable? 1

(a) cutting without planting
(b) planting a tree for every tree cut
(c) wasting water
(d) burning forests

22. Which observation suggests that the Earth rotates? 1

(a) the Moon is round
(b) stars appear to move across the night sky
(c) planets are bright
(d) the Sun is hot

23. A scientist is convinced that her hypothesis is right, but the data shows otherwise. The correct action is to: 1

(a) revise the hypothesis
(b) change the readings
(c) repeat until the data matches
(d) discard the data

24. Which statement explains why camels have broad feet? 1

(a) to climb
(b) to dig for water
(c) to avoid sinking in sand
(d) to swim

25. A child gets tired quickly, has pale skin and breathlessness. Which food should the doctor suggest? 1

(a) chips
(b) spinach and jaggery
(c) sugar
(d) butter

26. Two magnets, each with poles marked, are placed in a line with N of A facing S of B and they attract. If B is reversed, they will: 1

(a) lose their magnetism
(b) repel
(c) attract more strongly
(d) attract as before

27. A wheel of circumference 2.5 m makes 480 turns. Distance moved is: 1

(a) 2400 m **(b)** 1200 m **(c)** 120 m **(d)** 480 m

28. An iron cube and a wooden cube of the same size are placed in water. The iron sinks and the wood floats. What does this show? 1

(a) wood is magnetic
(b) wood is heavy
(c) iron is soluble
(d) iron is heavier for the same size

29. Which conclusion follows if the temperature of an object is not changing though heat is flowing in and out? 1

(a) object is cold **(b)** no heat flow
(c) object is hot **(d)** heat in equals heat out

30. Which factor does not affect the rate of evaporation? 1

(a) wind **(b)** area exposed
(c) temperature **(d)** the colour of the container

31. Which of these will take the longest to settle in water? 1

(a) pebbles **(b)** sand **(c)** gravel **(d)** fine clay

32. Which pair of observations would suggest that a dry-looking object is a living seed? 1

(a) it is brown and small **(b)** it swells and sprouts in water
(c) it is hard and dry **(d)** it breaks when pressed

33. Which of the following correctly shows the relative abundance in air? 1

(a) oxygen > nitrogen > carbon dioxide **(b)** carbon dioxide > oxygen > nitrogen
(c) nitrogen = oxygen **(d)** nitrogen > oxygen > carbon dioxide

34. If the new moon occurs on 1 March, the next new moon will be about: 1

(a) 8 March **(b)** 30 March **(c)** 22 March **(d)** 15 March

35. In the above experiment, which variable is measured as the outcome? 1

(a) growth of the plant **(b)** sunlight
(c) soil **(d)** amount of water

36. Which behaviour is an adaptation to a hot desert? 1

(a) being active at noon **(b)** being active at night
(c) hibernating in snow **(d)** swimming

37. Which combination in a vegetarian diet gives good-quality protein? 1

(a) sugar with butter **(b)** only fruits
(c) cereals with pulses **(d)** water with salt

38. Why does a steel almirah become slightly magnetic over many years? 1

(a) it is kept dry **(b)** Earth's magnetism slowly magnetises it
(c) it is exposed to sunlight **(d)** it is painted

39. Which of the following shows both rotational and linear motion at the same time? 1

(a) a swing **(b)** a pendulum **(c)** a fan **(d)** a rolling ball

40. In an experiment, a mixture of sand, salt and iron filings is given. The first step of separation should be: 1

(a) filtering **(b)** using a magnet **(c)** heating **(d)** adding water

41. **Assertion (A):** Sunlight is a renewable resource. 1
Reason (R): It is available every day and does not get used up by our use.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** The Sun is a star. 1
Reason (R): The Sun produces its own light and heat.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** Observation is the first step of the scientific method. 1
Reason (R): A telescope helps us observe objects that are very far away in the sky.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** Fish have a streamlined body shape. 1
Reason (R): A streamlined body helps a fish to move easily through water.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** Lack of vitamin C causes scurvy. 1
Reason (R): Citrus fruits like lemon and amla are rich in vitamin C.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** An iron nail is attracted by a magnet. 1
Reason (R): Cobalt and nickel are also magnetic materials.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** Standard units are needed to measure length. 1
Reason (R): Handspan and footstep differ from person to person.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** Wood is an opaque material. 1
Reason (R): Frosted glass is a translucent material.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** A clinical thermometer is used to measure the temperature of a human body. 1
Reason (R): A laboratory thermometer is used to measure the temperature of boiling water.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** Wet clothes dry faster on a windy day. 1
Reason (R): Moving air carries away water vapour quickly, so more water can evaporate.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**